Google Colab Lab Assignment -Pretrained Modle

Course Name: DEEP LEARNING LAB (MDM)

Lab Title: Research Paper Implementation with Pre-trained Model(ResNet50 for Image Classification)

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Research Paper Study and Implementation

Instructions:

1. Identify a research paper that utilizes a pre-trained model for a specific task.
2. Study the methodology, dataset, and model used in the research paper.
3. Implement the approach described in the research paper using the pre-trained model mentioned.
4. Compare your implementation results with the findings from the research paper.

Objective

1. Study a research paper utilizing a pre-trained model.
2. Reproduce the model implementation using the dataset and methodology from the research paper.
3. Fine-tune the pre-trained model and optimize hyperparameters.
4. Evaluate and compare model performance with the original research paper results.

Task 1: Research Paper Selection and Dataset Preparation (2 hours)

Instructions:

1. Select a research paper that applies a pre-trained model (e.g., VGG, ResNet, EfficientNet, etc.).
2. Identify the dataset used in the research paper and obtain or create a similar dataset.(**Mention Dataset Link and Description**)
3. Perform necessary preprocessing steps:

Resize images to match the model input dimensions.

Apply data augmentation techniques if applicable.

4. Split the dataset into training, validation, and testing sets.

Selected Research Paper

Title: "Transfer Learning Using Deep Convolutional Neural Networks for Image Classification"

Model Used: ResNet50

Technique: Transfer Learning + Fine-Tuning

Dataset Selected

Dataset: CIFAR-10 Dataset

Description

•60,000 images

•10 classes (airplane, car, bird, cat, etc.)

•32×32 images → resized to 224×224

```
# Task 1: Dataset Preparation

import tensorflow as tf

# Load dataset (online)
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()

# Reduce dataset size (IMPORTANT to avoid crash)
x_train = x_train[:10000]
y_train = y_train[:10000]

# Normalize
x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0

# Resize (smaller to save RAM)
x_train = tf.image.resize(x_train, (128,128)).numpy()
x_test = tf.image.resize(x_test, (128,128)).numpy()

# Validation split
x_val = x_train[:2000]
y_val = y_train[:2000]
```

```
x_train = x_train[2000:]
y_train = y_train[2000:]

print("Train:", x_train.shape)
print("Validation:", x_val.shape)
print("Test:", x_test.shape)
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 15s 0us/step
Train: (8000, 128, 128, 3)
Validation: (2000, 128, 128, 3)
Test: (10000, 128, 128, 3)
```

Task 2: Model Implementation and Fine-tuning

Instructions:

1. Implement the pre-trained model as described in the research paper.
2. Visualize feature maps of few layers
3. Freeze initial layers and fine-tune the top layers according to the paper's methodology.
4. Optimize hyperparameters such as:

Learning rate

Batch size

Number of epochs

Optimizer choice (Adam, SGD, RMSprop, etc.)

4. Document any modifications or enhancements made to improve performance.

```
# Task 2: Model (Transfer Learning - ResNet50)

from tensorflow.keras.applications import ResNet50
from tensorflow.keras import layers, models

# Load pretrained model
base_model = ResNet50(
    weights='imagenet',
    include_top=False,
    input_shape=(128,128,3)
)

# Freeze layers
for layer in base_model.layers:
    layer.trainable = False
```

```
# Add custom layers
x = base_model.output
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dense(128, activation='relu')(x)
x = layers.Dropout(0.5)(x)
output = layers.Dense(10, activation='softmax')(x)

model = models.Model(inputs=base_model.input, outputs=output)

# Compile
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

model.summary()
```

conv2_block1_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_1_b…
conv2_block1_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block1_1_r…
conv2_block1_2_bn (BatchNormalizatio…	(None, 32, 32, 64)	256	conv2_block1_2_c…
conv2_block1_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_2_b…
conv2_block1_0_conv (Conv2D)	(None, 32, 32, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block1_2_r…
conv2_block1_0_bn (BatchNormalizatio…	(None, 32, 32, 256)	1,024	conv2_block1_0_c…
conv2_block1_3_bn (BatchNormalizatio…	(None, 32, 32, 256)	1,024	conv2_block1_3_c…

conv2_block1_add (Add)	(None, 32, 32, 256)	0	conv2_block1_0_b… conv2_block1_3_b…
conv2_block1_out (Activation)	(None, 32, 32, 256)	0	conv2_block1_add…
conv2_block2_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block1_out…
conv2_block2_1_bn (BatchNormalizatio…	(None, 32, 32, 64)	256	conv2_block2_1_c…
conv2_block2_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_1_b…
conv2_block2_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block2_1_r…
conv2_block2_2_bn (BatchNormalizatio…	(None, 32, 32, 64)	256	conv2_block2_2_c…
conv2_block2_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_2_b…
conv2_block2_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block2_2_r…
conv2_block2_3_bn (BatchNormalizatio…	(None, 32, 32, 256)	1,024	conv2_block2_3_c…
conv2_block2_add (Add)	(None, 32, 32, 256)	0	conv2_block1_out… conv2_block2_3_b…
conv2_block2_out (Activation)	(None, 32, 32, 256)	0	conv2_block2_add…
conv2_block3_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block2_out…
conv2_block3_1_bn (BatchNormalizatio…	(None, 32, 32, 64)	256	conv2_block3_1_c…
conv2_block3_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_1_b…
conv2_block3_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block3_1_r…
conv2_block3_2_bn (BatchNormalizatio…	(None, 32, 32, 64)	256	conv2_block3_2_c…
conv2_block3_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_2_b…
conv2_block3_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block3_2_r…
conv2_block3_3_bn	(None, 32, 32,	1,024	conv2_block3_3_c…

(BatchNormalizatio…	256)		
conv2_block3_add (Add)	(None, 32, 32, 256)	0	conv2_block2_out… conv2_block3_3_b…
conv2_block3_out (Activation)	(None, 32, 32, 256)	0	conv2_block3_add…
conv3_block1_1_conv (Conv2D)	(None, 16, 16, 128)	32,896	conv2_block3_out…
conv3_block1_1_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block1_1_c…
conv3_block1_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_1_b…
conv3_block1_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block1_1_r…
conv3_block1_2_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block1_2_c…
conv3_block1_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_2_b…
conv3_block1_0_conv (Conv2D)	(None, 16, 16, 512)	131,584	conv2_block3_out…
conv3_block1_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block1_2_r…
conv3_block1_0_bn (BatchNormalizatio…	(None, 16, 16, 512)	2,048	conv3_block1_0_c…
conv3_block1_3_bn (BatchNormalizatio…	(None, 16, 16, 512)	2,048	conv3_block1_3_c…
conv3_block1_add (Add)	(None, 16, 16, 512)	0	conv3_block1_0_b… conv3_block1_3_b…
conv3_block1_out (Activation)	(None, 16, 16, 512)	0	conv3_block1_add…
conv3_block2_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block1_out…
conv3_block2_1_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block2_1_c…
conv3_block2_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_1_b…
conv3_block2_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block2_1_r…
conv3_block2_2_bn (BatchNormalizatio	(None, 16, 16, 128)	512	conv3_block2_2_c…

(BatchNormalizatio…	128)		
conv3_block2_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_2_b…
conv3_block2_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block2_2_r…
conv3_block2_3_bn (BatchNormalizatio…	(None, 16, 16, 512)	2,048	conv3_block2_3_c…
conv3_block2_add (Add)	(None, 16, 16, 512)	0	conv3_block1_out… conv3_block2_3_b…
conv3_block2_out (Activation)	(None, 16, 16, 512)	0	conv3_block2_add…
conv3_block3_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block2_out…
conv3_block3_1_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block3_1_c…
conv3_block3_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_1_b…
conv3_block3_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block3_1_r…
conv3_block3_2_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block3_2_c…
conv3_block3_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_2_b…
conv3_block3_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block3_2_r…
conv3_block3_3_bn (BatchNormalizatio…	(None, 16, 16, 512)	2,048	conv3_block3_3_c…
conv3_block3_add (Add)	(None, 16, 16, 512)	0	conv3_block2_out… conv3_block3_3_b…
conv3_block3_out (Activation)	(None, 16, 16, 512)	0	conv3_block3_add…
conv3_block4_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block3_out…
conv3_block4_1_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block4_1_c…
conv3_block4_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_1_b…
conv3_block4_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block4_1_r…

conv3_block4_2_bn (BatchNormalizatio…	(None, 16, 16, 128)	512	conv3_block4_2_c…
conv3_block4_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_2_b…
conv3_block4_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block4_2_r…
conv3_block4_3_bn (BatchNormalizatio…	(None, 16, 16, 512)	2,048	conv3_block4_3_c…
conv3_block4_add (Add)	(None, 16, 16, 512)	0	conv3_block3_out… conv3_block4_3_b…
conv3_block4_out (Activation)	(None, 16, 16, 512)	0	conv3_block4_add…
conv4_block1_1_conv (Conv2D)	(None, 8, 8, 256)	131,328	conv3_block4_out…
conv4_block1_1_bn (BatchNormalizatio…	(None, 8, 8, 256)	1,024	conv4_block1_1_c…
conv4_block1_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_1_b…
conv4_block1_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block1_1_r…
conv4_block1_2_bn (BatchNormalizatio…	(None, 8, 8, 256)	1,024	conv4_block1_2_c…
conv4_block1_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_2_b…
conv4_block1_0_conv (Conv2D)	(None, 8, 8, 1024)	525,312	conv3_block4_out…
conv4_block1_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block1_2_r…
conv4_block1_0_bn (BatchNormalizatio…	(None, 8, 8, 1024)	4,096	conv4_block1_0_c…
conv4_block1_3_bn (BatchNormalizatio…	(None, 8, 8, 1024)	4,096	conv4_block1_3_c…
conv4_block1_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_0_b… conv4_block1_3_b…
conv4_block1_out (Activation)	(None, 8, 8, 1024)	0	conv4_block1_add…
conv4_block2_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block1_out…